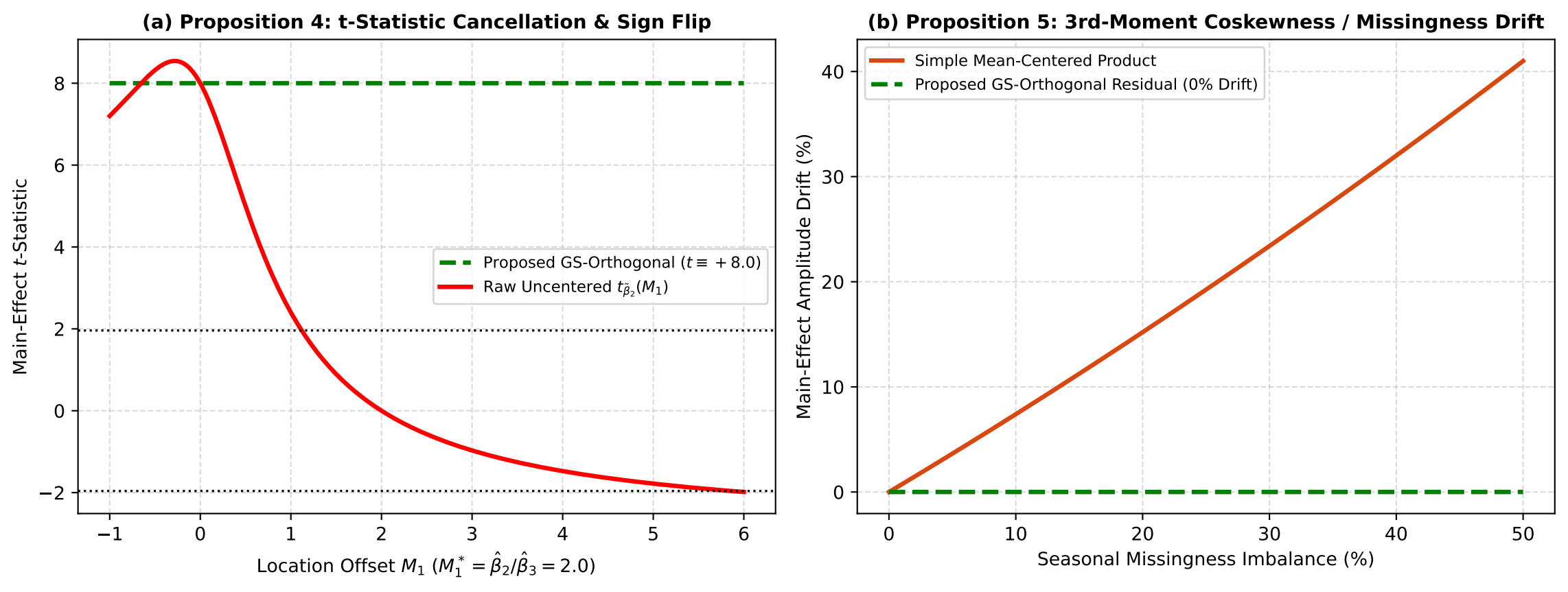
(a) Proposition 4: t-Statistic Cancellation & Sign Flip
Main-Effect t-Statistic
Location Offset M_1 ($M_1^* = \hat{\beta}_2/\hat{\beta}_3 = 2.0$)
Proposed GS-Orthogonal ($t \equiv +8.0$)
Raw Uncentered $t_{\tilde{\beta}_2}(M_1)$
8
6
4
2
0
−2
−1
0
1
2
3
4
5
6
(b) Proposition 5: 3rd-Moment Coskewness / Missingness Drift
Main-Effect Amplitude Drift (%)
Seasonal Missingness Imbalance (%)
Simple Mean-Centered Product
Proposed GS-Orthogonal Residual (0% Drift)
40
30
20
10
0
0
10
20
30
40
50